

RDW28801 series intergrated 18.0KW

Vehicle DC-DC Converter

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Product Introduction

The **RDW28801** series vehicle-mounted DC-DC converter is a high-power , high-density, and high-efficiency DC-DC converter specifically developed for new energy vehicle models such as new lithium-ion electric vehicles, logistics vehicles, special vehicles, and construction machinery . It is designed and developed using a modular, standardized, and universal design approach .

The module is designed with fully digital control technology, offering flexible and intelligent control, excellent protection features, and strong system robustness . It has an integrated microprocessor that communicates with the monitoring unit, and internal parameters can be set or adjusted by the upper-level monitoring unit via the CAN interface.

multiple protection functions, including input over/under voltage protection, output overcurrent protection, output overvoltage protection, output short circuit protection, and overtemperature protection .

Main specifications:

model	Input voltage range	Rated output power	Rated output voltage	Output voltage/current range	3D model
RDW28871-01	400-900VDC	18.0KW	27VDC	0-32VDC/0-660A	RDW28801 V1.4 .stp
RDW28871-02	500-1100VDC	18.0KW	27VDC	0-32VDC/0-660A	
RDW28881-01	400-900VDC	18.0KW	54VDC	0-68VDC/0-333A	
RDW28891-01	300-750VDC	6.0KW	72VDC	0-95V/0-240A	

1. Electrical characteristics

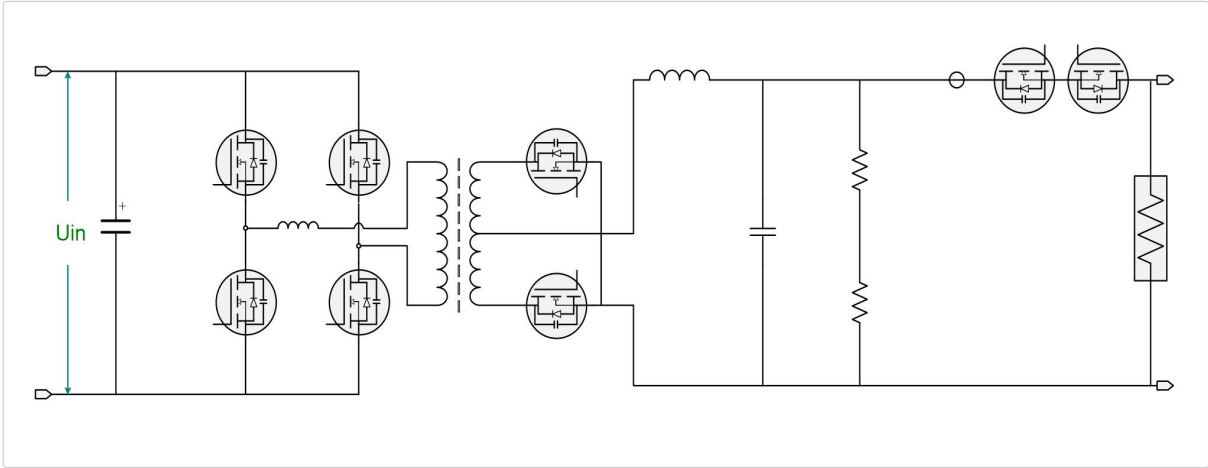
1.1. Electrical characteristics

model			
Vehicle power supply type	Air-cooled automotive DC-DC converter		
Model and supplier part number	RDW28871-01	RDW28871-02	RDW28881-01
Input characteristics			
Rated input voltage	540V	720V	540V
Input voltage range	400 ~ 900V	500 ~ 1100V	400 ~ 900V
Input precharge circuit	built-in		

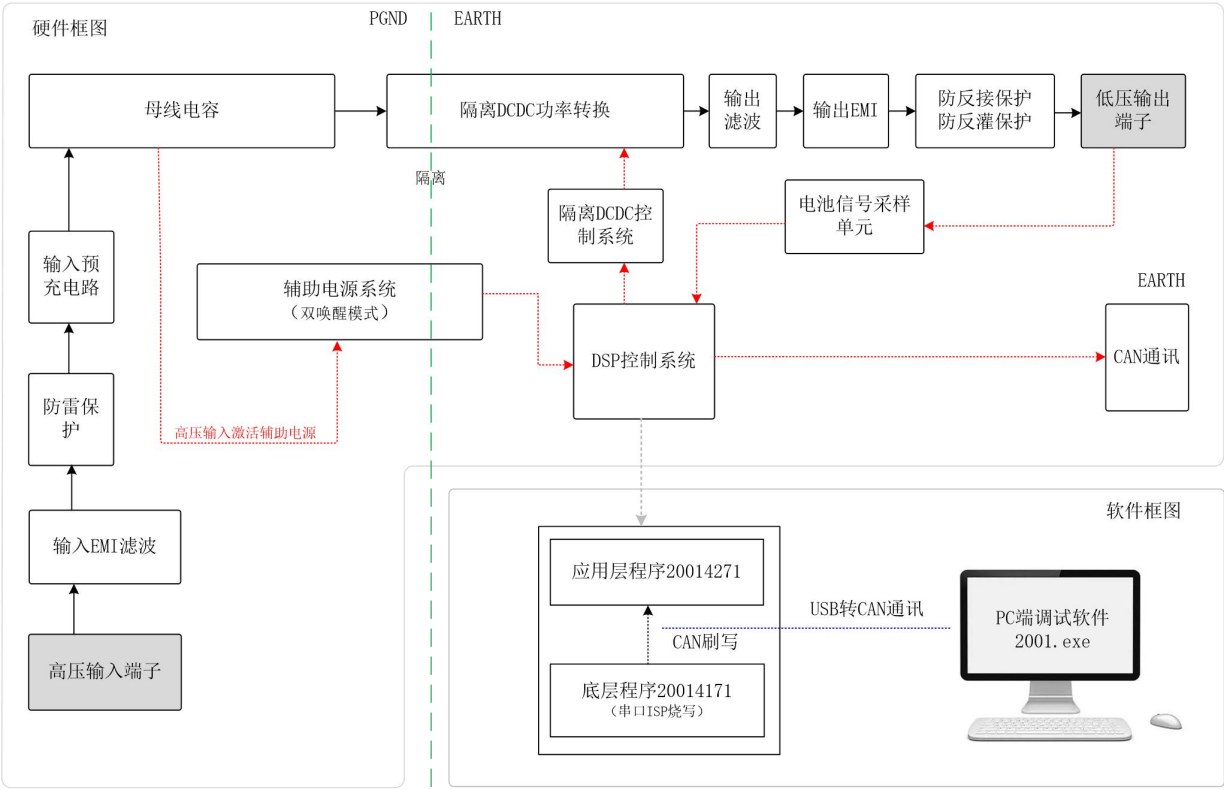
Initiation inrush current	≤33A	≤33A	≤33A
bus capacitor	36uF	36uF	36uF
Output characteristics			
Rated output power	18.0KW	18.0KW	18.0KW
Rated output voltage	27V	27V	54V
Output voltage range	0 ~ 32V	0 ~ 32V	0 ~ 68V
Output current range	0 ~ 660A	0 ~ 660A	0 ~ 333A
Voltage regulation accuracy (test at the root of the lead wire)	±2%	±2%	±2%
Output response time	≤200mS		
Typical efficiency	≥92%	≥92%	≥93%
Peak power	TBD	TBD	TBD
Boot time	<1s		
Operating noise	-		
Protective features			
Over/under voltage protection	The system can automatically recover from shutdown due to input overvoltage or undervoltage , and the system can automatically recover from shutdown due to output overvoltage or undervoltage .		
Output reverse connection and short circuit protection	when output is reversed , but self-recovering. Output short-circuit protection (hiccups, self-recovering when the fault is removed).		
Over-temperature protection	When the heat sink temperature exceeds 75℃, the output power is reduced; when the temperature exceeds 95℃, the circuit is disconnected; the charger resumes output power when the charging temperature returns to below 85℃.		
Environmental conditions			
Operating ambient temperature	-40℃~+85℃		
Storage temperature	-40~95℃		
humidity	5% - 95% No condensation, no dew		
IP Level	IP67		
Cooling function	air-cooled		
Communication function	CAN bus		
Control method	No enable control; Hard-wired high-level enable control; CAN communication enable control		
Safety characteristics			

Dielectric strength	Primary-Secondary Side 2000VAC	Original edge - casing 1500VAC
Insulation resistance	Primary side - secondary side $\geq 50M\Omega$	
Vibration resistance	After frequency sweep vibration tests in the X, Y, and Z directions, no damage was found to the components, and no fasteners were found to be loose.	
Impact resistance	See requirement 6.5 in GB/T15139-1994.	
Resistance to industrial solvents	Metal parts have a good anti-corrosion coating	
Salt spray resistance	See GB/T 2423.17	
Durability	In accordance with the relevant provisions of GB/T 24347-2009	
EMC characteristics		
Electromagnetic immunity	It meets the requirements of Chapter 4 of GB/T17619-1998.	
Electromagnetic interference	See the limits specified in Chapters 12 and 14 of GB18655-2002.	

1.1. Electrical topology diagram



1.2. System block diagram

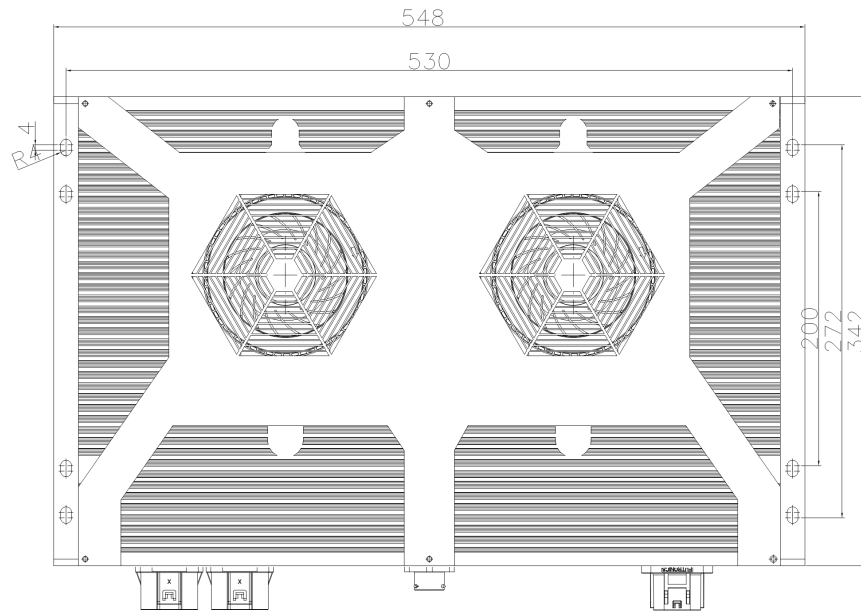


2. External dimensions and weight

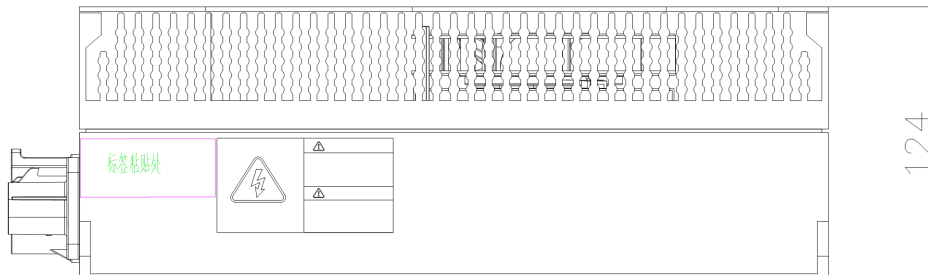
2.1. Product Dimensions

3D model data: RDW28801 V1.2

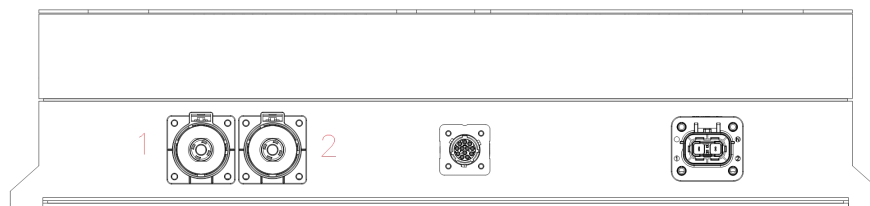
Top view



Side view



Front view

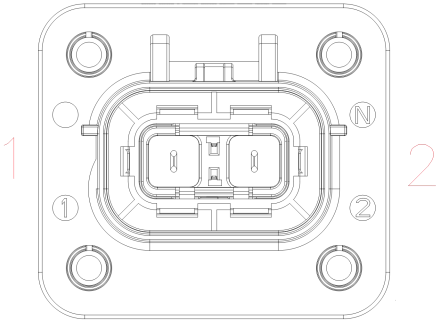
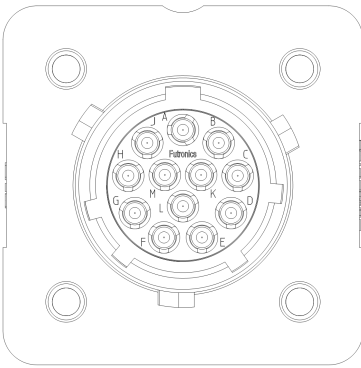
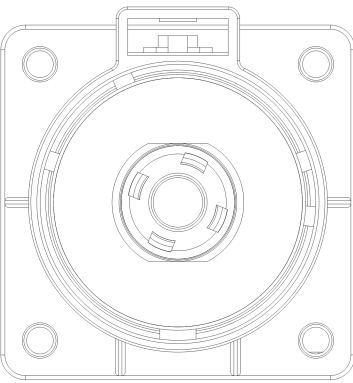


2.2. Product weight

31 kg ± 0.5 kg

3. Connector and terminal definition

3.1. Connector Models and Definitions

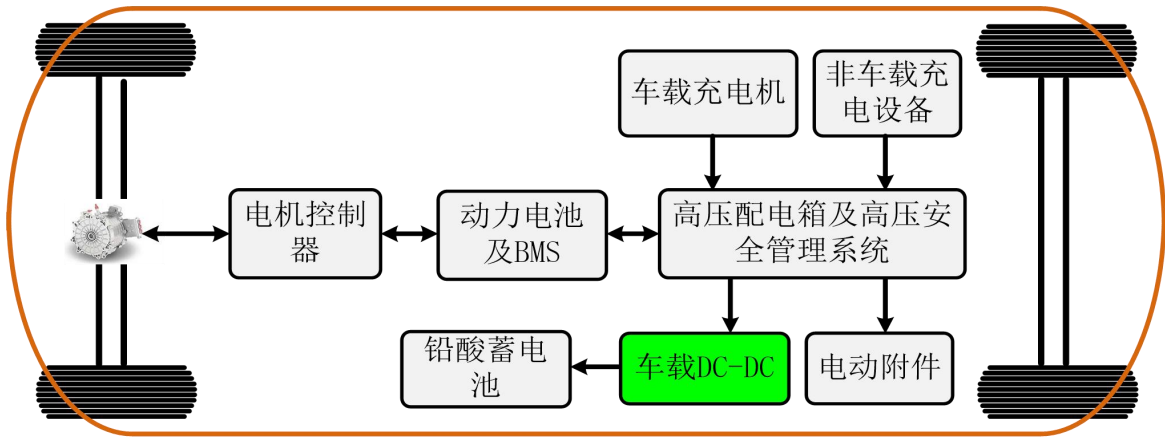
type	Connector definition		Connector drawings
enter FHV200201TZ-04S01	foot posit ion	definition	
	1	Input +	
	2	enter-	
	Connector manufacturers		/
	For plug-in models		FHV600201TZ-04S01
Signal FPT20N21212APN	foot posit ion	definition	
	1	C ANH	
	2	C ANL	
	3	ENABLE	
	4	NG	
	5	NG	
	Connector manufacturers		/
	For plug-in models		FPT60021212ASN
Output PRC00X-500-10M10	foot posit ion	definition	
	red	Output positive terminal	
	black	Output negative terminal	
	Connector manufacturers		/
	For plug-in models		PRC48X-500-120

3.2. Signal interface schematic diagram

Signal Description	
CAN communication	
Fan drive	

4. User Guide

4.1. Electrical connection diagram



4.2. Product Installation

Product Part Number	RDW28801 series	
Product Type	Vehicle-mounted DC-DC unit	
Mounting screws	Mounting hole diameter	Φ 8
	Number	8
	Screw model recommendations	M 8 *10 socket head cap screws

Install and fix this product

Align the mounting holes, tighten the screws, and secure the power supply.

Tightening torque requirements

During installation, apply appropriate torque based on screw size, connection method, etc.

See the table below for details:

Specifications and Models		Tightening torque (torque range: ±10%) / (unit: Kgf.cm)						
Major Categories	Subcategories	Plastics	Steel-Plastic Copper-copper	General connection		High-density connectivity		
				Steel-steel	Copper-cast aluminum Steel-aluminum profiles Steel-copper	Steel-steel	Steel-cast aluminum Steel-Copper	Steel-aluminum profiles
Hex socket screws	M2		0.8	1.5	1.5	2.5	2.5	1.5
	M2.5		1.6	3	3	5.5	4.5	3
	M3	1.5	3	5.5	5	10	8	6
	M4		6	12	10	16	14	12
	M5		10	20	13	30	28	20

	M6		15	30	28	50	48	30
	M8					80	80	—

4.3. CAN communication protocol

project	Technical indicators	Remark
Crystal Oscillator Tolerance	$\pm 0.15\%$	Within the operating temperature range
Communication rate	The configuration can be set via the backend software to ensure data retention even after power failure.	Tolerance is ± 0.375 Kbit/s
Sampling points	The sampling point should be set at approximately 7/8 of the bit time, but no later than 7/8 of the bit time.	
transceiver	The largest transceiver "loop delay" (from send to receive) is 300 ns.	The CAN transceiver should conform to the ISO 11898-2 standard.
Termination resistor	DC-DC CAN communication circuits do not have a 120-ohm terminating resistor by default.	
Default CAN communication protocol	 车载DCDC默认CAN控制通讯协议 V1.	

4.4. Background debugging software instructions

Product Part Number	RDW28801 series	
Product Type	Vehicle-mounted DC-DC unit	
Backend software coding	2001 Setup V2.0	
Backend software communication methods	CAN communication	Baud rate adjustable 125K/250K/500K
Installation and Usage Help		 上位机使用说明.pdf
Support CAN box Brand 1	1. Beijing Aite USBCAN-2I 2. Beijing Aite USBCAN-I	 USBCAN Driver for Windows 10-amd64-1.0.1.exe
Support CAN box Brand 2	TBD	

4.5. Troubleshooting and Confirmation

Fault	Common causes of failure	troubleshooting
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phenomenon		
No power output	High voltage input abnormal (no voltage or reversed connection)	Check if the high voltage input is normal.
	12V voltage input port malfunction (no voltage, over/under voltage, reverse connection)	Check if the 12V voltage input port is normal.
	Output circuit break	Check if the output connection is normal.
DC-DC no message	The signal connector is not properly connected.	Re-plug the signal connector
	CAN wire reversed	Adjust CAN bus wiring sequence
	Communication protocol mismatch	Compare whether the protocols match.
	baud rate mismatch	Compare whether the baud rates match.
High-voltage input fuse in distribution box damaged Product reports fault signal	Input short circuit	Check if the high voltage input is normal.
	Input over/under voltage, output over/under voltage, over temperature, output short circuit/over current	Check the input voltage and output for overcurrent/short circuit. Turn off the power and let it sit for 10 minutes. If the problem persists, please contact the manufacturer.
Over-temperature fault	Air-cooled machines: Fan stalled or air duct blocked	Check the fan and air duct
	Water-cooled machines: No coolant or coolant temperature too high	Check if the coolant level is normal.

5. User Guidelines and Precautions

Please read the warnings and precautions before using the product. Improper operation may result in electric shock or fire. Please ensure you have read the warnings and precautions before using the product.

warn:

Unauthorized disassembly of the product for repair, debugging, or modification is strictly prohibited.

When the product is powered on, please keep your hands and face away from it to avoid accidental injury.

The product contains high pressure and high temperature. Do not touch the internal components, as this may cause electric shock or burns.

If there are any unusual noises or smells from the power supply during use, please turn off the input immediately.

Connectors that meet specifications must be used to ensure that all plugs and sockets are securely connected. Loose connections may cause localized overheating and fire.

Please use the power supply within the technical specifications. Using it outside the specifications may damage the product.

Please avoid leaving the product in a rainy location for extended periods;

Before installation, please ensure that the casing is intact. If there is any damage,

please replace it immediately or contact the manufacturer.

Precautions:

Confirm that the product input/output terminals and signal terminals are connected correctly according to the product manual; when wiring, please disconnect the input power and never plug or unplug connectors while the power is on.

This power input/output terminal requires an external fuse or other overcurrent protection device;

The potential electrical hazards at the output end during product use must be considered to ensure that end-users do not come into contact with the product; terminal equipment manufacturers must design appropriate protection schemes to ensure that no danger arises during operation due to accidental contact of power terminals by engineers or tools.

If the equipment's safety protection is compromised, the equipment must be stopped and handled in accordance with the relevant maintenance regulations.

When power equipment is transferred from a cold environment to a warm environment, condensation may cause leakage hazards, so grounding requirements must be strictly implemented;

Only qualified personnel can connect the equipment to the power source.

Power must be cut off and the machine stopped for five minutes to allow the capacitors sufficient time to discharge before any maintenance or repairs can be performed on the power supply equipment.

Safety precautions: Avoid touching areas marked with safety warning signs or high voltage signs to prevent electric shock or burns.

6. Referenced standards and specifications

GB 14023-2011 Limits and methods of measurement for radio interference characteristics of vehicles, ships and devices driven by internal combustion engines

GB/T 17626.2-2006 Electromagnetic compatibility testing and measurement techniques – Electrostatic discharge immunity test

GB/T 17626.3-2006 Electromagnetic compatibility testing and measurement techniques – Radio frequency electromagnetic field radiated immunity test

GB/T 17626.4-2008 Electromagnetic compatibility testing and measurement techniques – Electrical fast transient/burst immunity test

GB/T 17626.5-2008 Electromagnetic compatibility testing and measurement techniques – Surge (impulse) immunity test

GB/T 17619-1998 Limits and methods for measuring electromagnetic radiation immunity of electronic and electrical components in motor vehicles

GB/T 18384.3-2015 Safety requirements for electric vehicles – Part 3: Protection against electric shock

GB/T 18387-2008 Limits and measurement methods for electromagnetic field emission intensity of electric vehicles, broadband, 9kHz~30MHz

GB/T 18487.2-2001 Electric Vehicle Conductive Charging System – Connection Requirements between Electric Vehicles and AC/DC Power Sources (doc)

GB/T 18487.3-2001 Conductive Charging System for Electric Vehicles – AC/DC Chargers (Stations) for Electric Vehicles (doc)

GB/T 18488.1-2015 Drive motor systems for electric vehicles – Part 1: Technical requirements

GB/T 18655-2010 Limits and methods of measurement for radio interference characteristics of marine and internal combustion engines for the protection of vehicle-mounted receivers

GB/T 19826-2014 General Technical Conditions and Safety Requirements for DC Power Supply Equipment in Power Engineering

GB/T 21437.2-2008 Electrical disturbances caused by conduction and coupling in road vehicles—Part 2: Electrical transients conducted along power lines

GB/T 2423.1-2008 Environmental testing of electrical and electronic products – Part 2: Test methods – Test A: Low temperature

GB/T 2423.2-2008 Environmental testing of electrical and electronic products – Part 2: Test methods – Test B: High temperature

GB/T 2423.3-2006 Basic Environmental Testing Procedures for Electrical and Electronic Products—Test Ca: Constant Damp Heat Test Method;

GB/T 2423.4-2008 Basic Environmental Testing Procedures for Electrical and Electronic Products—Test Db: Alternating Damp Heat Test Method

GB/T 2423.5-1995 Environmental testing of electrical and electronic products—Part 2: Test methods/Test Ea and guidelines: Impact

GB/T 2423.6-1995 Environmental testing of electrical and electronic products—Part 2: Test methods/Test Ea and guidelines: Impact

GB/T 2423.8-1995 Environmental testing of electrical and electronic products—Part 2: Test methods/Test Ed: Free fall

GB/T 2423.10-2008 Environmental testing of electrical and electronic products—Part 2: Test methods/Test Fc and guidelines: Vibration (sine wave)

GB/T 2423.22-2012 Environmental testing of electrical and electronic products—Part 2: Test

N: Temperature change

GB/T 24347-2009 DC/DC Converters for Electric Vehicles

GB 4208-2008 Degrees of protection provided by enclosures (IP codes)

QC/T 413-2002 Basic Technical Requirements for Automotive Electrical Equipment

GB 9254-2008 Limits and methods of measurement for radio interference from information technology equipment

7. Packaging, transportation, and storage

Package

The product packaging information is as follows:

Packaging quantity and box information	Single unit net weight (kg)	31 kg ± 0.5 kg
	Packaging box outer dimensions (mm)	390*295*157
	Quantity of a single unit	1
	Total weight after packaging (kg)	35 kg ± 0.5 kg

The packaging box should display the product name, product model, and manufacturer’s name; the technical documents supplied with the product in the packaging box should include the product’s certificate of conformity.

Products should be transported in sturdy packaging boxes, with the boxes conforming to relevant national standards and bearing markings such as “Handle with Care” and “Keep Dry.” Packaging boxes containing products are permitted to be transported using various means of transport. Direct exposure to rain and snow, as well as mechanical impact, should be avoided during transport. Shipping markings should be affixed, as shown in Figure 7-2 below.



Transport signs

Storage

When the product is not in use, it should be stored in its packaging box. The warehouse environment temperature should be between -10 and 40° C and the relative humidity should not exceed 80%. The warehouse should not contain harmful gases, flammable or explosive products, or corrosive chemicals, and should be free from strong mechanical vibration, impact, and strong magnetic fields. The packaging box should be raised at least 20cm off the ground and at least 50cm away from walls, heat sources, windows, or air inlets. The storage period under these conditions is generally 2 years. After 2 years, the product should be re-inspected.

The product should be stored in a well-ventilated, dry place. It must be kept away from high temperatures, fire sources, and chemicals. Store neatly and avoid throwing or dropping.

8. Version update log

date	Version	Reason for change	Remark
2021/03/13	V1.0		
2021/11/07	V1.1	Update plugins	
2022/12/12	V1.2	Update the casing	
2024/05/06	V1.3	Update the casing	
2026/03/02	V1.4	Update content	